

Business Plan

Aegis Metrology & Automation LLC

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Proposed Company Name: Aegis Metrology & Automation LLC
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Name Rationale: *"Aegis"* denotes protection, trust, and validation—reflecting specialized focus on failure analysis, defect localization, and fault verification[cite: 1]. Combined with *"Metrology & Automation,"* the brand clearly communicates high-value technical precision to semiconductor manufacturers, government research entities, and defense primes across the Pacific Northwest.

1 EXECUTIVE SUMMARY

1.1 1.1 Company Overview

Aegis Metrology & Automation LLC is a Portland-based deep-tech research and engineering consulting firm specializing in automated defect localization, machine vision inspection, and software-driven operational automation for advanced microelectronics and scientific infrastructure[cite: 1].

1.2 1.2 Mission Statement

To accelerate semiconductor failure analysis, optimize laboratory workflows, and build inclusive STEM workforce pipelines by uniting physical device characterization with advanced machine learning[cite: 1, 2].

1.3 1.3 Core Business Model

Operating as a zero-capital, non-dilutive business startup, Aegis Metrology & Automation leverages a three-tiered business structure:

1. **Federal & State R&D Grants:** Securing NSF, DoD, and Business Oregon grants through Small Business Technology Transfer (STTR) partnerships[cite: 2].
2. **B2B Engineering & Automation Consulting:** Delivering high-margin software development, process tool automation, and workflow integration for commercial tool vendors and semiconductor labs[cite: 1].
3. **Workforce Development & Social Enterprise Grants:** Partnering with regional agencies to deliver specialized microelectronics and lab automation training programs[cite: 1, 2].

2 COMPANY & LEADERSHIP STRUCTURE

2.1 2.1 Corporate Structure

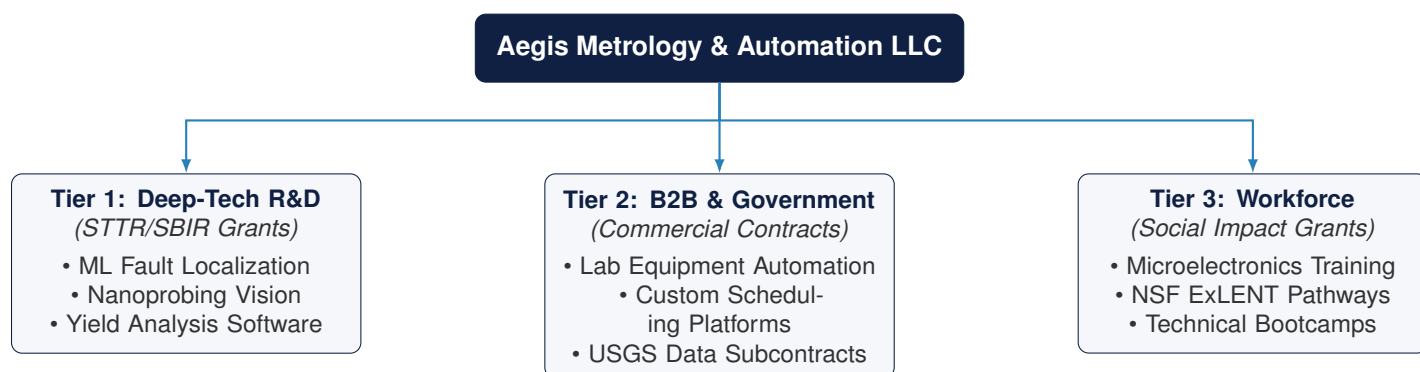
- **Legal Entity:** Limited Liability Company (LLC) registered in Oregon.
- **Ownership:** 100% Founder-Owned (Ryne Frank Shelton)[cite: 1].
- **Location:** Portland Metro Area (Hillsboro / Portland Innovation Corridor)[cite: 1].

2.2 2.2 Founder & Principal Engineer Profile

The company is led by Ryne Frank Shelton, an Electrical and Computer Engineer currently completing an M.S. in Electrical Engineering (4.00 GPA) at Portland State University[cite: 1]. The founder brings specialized, multi-disciplinary domain expertise:

- **Semiconductor Failure Analysis:** Hands-on experience at Intel Corporation conducting TEM, SEM, nanoprobeing, Electrical Failure Analysis (EFA), and Physical Failure Analysis (PFA) on advanced logic and SRAM structures[cite: 1].
- **Process Equipment & Automation:** Reduced equipment setup times by over 80% at Hitachi High-Tech through workflow automation; experienced in vacuum systems, dry-etch, RF systems, and gas delivery tools[cite: 1].
- **Machine Learning & Vision Systems:** Developed CNN/LSTM machine-vision models for optical inspection at Datalogic and anomaly detection models for geophysical datasets with the U.S. Geological Survey (USGS)[cite: 1].
- **Research & Leadership:** Served as Principal Investigator under the NASA Oregon Space Grant Consortium (OSGC) and managed Portland State University's Engineering Learning Center supervising 10–15 technical employees[cite: 1, 2].
- **Community Leadership:** Active involvement in STEM equity initiatives, including Louis Stokes Alliance for Minority Participation (LSAMP) and Mathematics, Engineering and Science Achievement (MESA) mentorship[cite: 2].

3 PRODUCTS & SERVICE OFFERINGS



3.1 3.1 Tier 1: R&D & IP Development (STTR/SBIR Funded)

- **Automated Defect Localization Engine (ADLE):** A software suite utilizing deep learning (CNNs and pattern recognition) to analyze nanoprobeing signals, SEM/TEM imaging outputs, and SRAM bitmapping to automate fault localization on advanced-node devices[cite: 1].

- **STTR Cleanroom Access Model:** Partners with Portland State University research facilities to access physical analytical tools without capital expenditure obligations[cite: 1, 2].

3.2 3.2 Tier 2: Commercial Consulting & Federal Subcontracting

- **Lab Process Automation:** Custom Python/MicroPython scripts, tool calibration workflows, and signal-path verification integrations for tool manufacturers and research labs[cite: 1].
- **Laboratory Asset & FOUP Management Software:** Cloud-enabled lab management tools building upon proven architecture from RYSYS[cite: 1].
- **Geophysical & Sensor Data Analytics:** Federal subcontracting support for agencies like the USGS, providing machine learning data pipelines and anomaly detection[cite: 1].

3.3 3.3 Tier 3: Workforce Development Programs

- **Semiconductor Technician Bootcamps:** Hands-on training modules covering cleanroom operations, vacuum/RF systems maintenance, and automation basics, targeting non-traditional and underrepresented engineering talent[cite: 1, 2].

4 MARKET & REGIONAL CONTEXT ANALYSIS

4.1 4.1 Target Market Opportunity (Portland "Silicon Forest")

The Portland metropolitan region represents one of North America's premier semiconductor manufacturing and R&D clusters, housing major facilities for Intel, Microchip Technology, Qorvo, Hitachi High-Tech, and Microbeam[cite: 1].

- **Primary Commercial Clients:** Tool vendors and testing laboratories requiring custom automation to increase throughput and reduce calibration setup times[cite: 1].
- **Federal R&D Funding Environment:** Federal backing under the CHIPS and Science Act prioritizing domestic failure analysis, yield optimization, and workforce development.

4.2 4.2 Competitive Advantage

- **Zero Overhead Structure:** Operating lean without owned physical lab space allows Aegis to price B2B consulting services competitively while focusing capital on research talent[cite: 1].
- **Rare Interdisciplinary Expertise:** Direct combination of physical device failure analysis (nanoprob-ing, TEM/SEM)[cite: 1] with production-grade machine learning and software development[cite: 1].
- **Established Institutional Relationships:** Strong existing ties with PSU academic departments, USGS research teams, and regional equity networks[cite: 1, 2].

5 NON-DILUTIVE CAPITAL & FUNDING STRATEGY

Because initial capital is \$0, operations rely on non-dilutive grant awards, state matching programs, and immediate consulting cash flows.

Funding Mechanism	Target Entity / Agency	Estimated Award	Strategic Objective
Business Oregon Phase 0	State of Oregon	\$5,000 – \$10,000	Fund proposal prep for federal submissions.
NSF STTR Phase I	National Science Foundation	\$275,000 – \$305,000	Develop proof-of-concept ML fault localization software[cite: 1].
Business Oregon Phase I Match	State of Oregon	\$50,000 – \$100,000	State matching funds following NSF Phase I.
NSF ExLENT Program	National Science Foundation	\$250,000 – \$1,000,000	Launch workforce training pathways[cite: 2].
Federal Subcontracts	USGS / DoD Primes	\$30,000 – \$80,000/proj	Data modeling & tool automation consulting[cite: 1].

Table 1: Grant and Subcontract Funding Pipeline

6 OPERATIONAL & FACILITY STRATEGY

• Phase I (Months 1–12): Virtual & Academic Partnership Model

- Business administration and software development hosted virtually[cite: 1].
- Physical device characterization and analytical research conducted via user agreements at Portland State University cleanrooms and analytical facilities[cite: 1, 2].
- Business registration through Oregon Secretary of State; federal procurement setup on SAM.gov.

• Phase II (Months 13–24): Collaborative Hub Expansion

- Secure small office footprint within a regional incubator (e.g., Beaverton/Hillsboro enterprise centers).
- Expand B2B software consulting to cover core operating overhead and hire initial junior software research staff.

7 FINANCIAL PLAN & 3-YEAR PROJECTIONS

7.1 Financial Model Assumptions

- \$0 personal equity capital required at start.
- Operating costs in Year 1 limited to business registration, software licensing, and cloud computing resources.
- Revenue driven by immediate B2B consulting engagements, followed by STTR Phase I grant disbursements[cite: 1].

7.2 7.2 Three-Year Revenue & Expense Forecast

Financial Metric	Year 1	Year 2	Year 3
Consulting & Contract Revenue	\$45,000	\$120,000	\$220,000
Grant Revenue (NSF / Business Oregon)	\$280,000	\$400,000	\$750,000
Total Revenue	\$325,000	\$520,000	\$970,000
Operating Expenses (SW, Cloud, Legal)	\$25,000	\$45,000	\$75,000
University Facility Leasing (PSU)	\$35,000	\$60,000	\$90,000
Personnel & Subcontractors	\$180,000	\$310,000	\$580,000
Net Pre-Tax Profit / Runway Retention	\$85,000	\$105,000	\$225,000

Table 2: Three-Year Revenue and Expense Forecast

8 IMPLEMENTATION ROADMAP & MILESTONES

1. Q1 2026: Foundation & Setup

- Register Aegis Metrology & Automation LLC in Oregon.
- Complete SAM.gov & eRA Commons registration.
- Apply for Business Oregon Phase 0 grant for proposal support.

2. Q2 2026: Commercial Outreach & STTR Proposal

- Pitch local tool vendors on B2B setup automation & scheduling software[cite: 1].
- Formalize STTR partnership agreement with PSU[cite: 1, 2].
- Submit NSF STTR Phase I proposal (Microelectronics/AI topic)[cite: 1].

3. Q3–Q4 2026: Initial Revenue & Contract Execution

- Execute 1–2 commercial B2B automation subcontracts[cite: 1].
- Submit workforce grant application under NSF ExLENT[cite: 2].
- Deliver USGS geophysical data analytics subcontract work[cite: 1].

4. 2027: STTR Execution & Scale

- Commence NSF Phase I grant execution upon award (\$275k+).
- Apply for Business Oregon Matching Grant (\$50k–\$100k).
- Expand technical training initiatives with MESA/LSAMP community partners[cite: 2].

9 RISK MANAGEMENT MATRIX

Identified Risk	Severity	Risk Mitigation Strategy
Delays in Grant Funding Approval	High	Rely on short-term B2B automation consulting contracts (e.g., setup time reduction, scheduling platforms) for immediate cash flow[cite: 1].
Lack of Owned Cleanroom Equipment	High	Execute formal STTR MOU with Portland State University to utilize existing TEM/SEM and nanoprobeing equipment[cite: 1, 2].
Commercial Client Acquisition Delays	Medium	Leverage MECOP industry contact networks, Hitachi High-Tech connections, and regional semiconductor conferences[cite: 1, 2].

Table 3: Risk Assessment and Mitigation Matrix